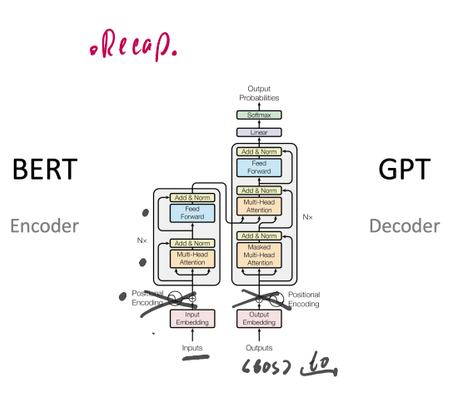
• Recap.


BERT
Encoder
GPT
Decoder
<bos> t0

• Relative embeddings / Relative position encoding

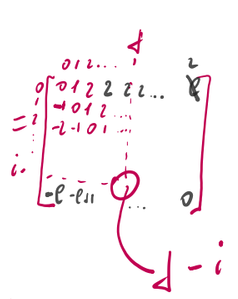
Ты должен верить в силу. →

В силу должен верить ты. →

$$\text{RelativeAttention} = \text{Softmax}\left(\frac{QK^\top + S_{rel}}{\sqrt{D_h}}\right)V$$

$x = [\ ]$; $xW^q = Q$; $xW^k = K$; $\underset{l\times d}{Q}\ \underset{d\times l}{K^T} = \underset{l\times l}{[\ ]}$;

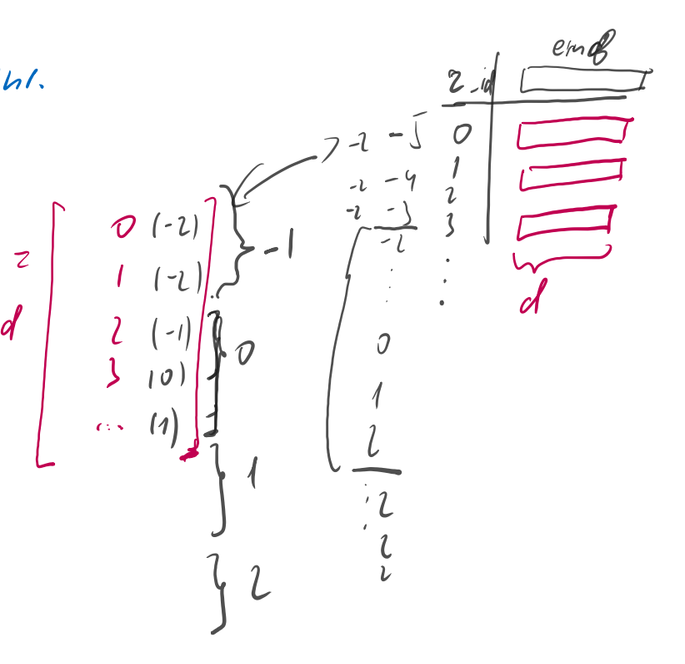
• Scalar-valued relative position encoding


S_{rel} =
0 1 2 ... j
0 1 2 2 2 ...
-1 0 1 2
-2 -1 0 1
-l -l+1
j - i

j-i	scalar_param
-5	☐
-4	☐
-3	☐
-2	☐
...	
0	☐
1	⋮
...	
3	
4	
5	

• Vector-valued rel. pos. enc.

$\to S_{rel} = \underset{l\times d}{Q} \cdot \underset{d\times l}{R^T}$; $\Rightarrow \underset{l\times d}{R} =$

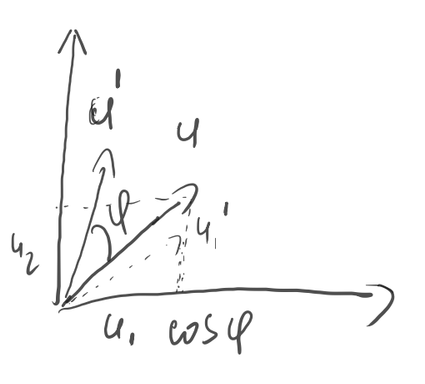

0 (-2)
1 (-2)
2 (-1)
3 (0)
... (1)
-1
0
1
2
j-i
emb
-5
-4
-3
0
1
2
3
d

• RoPE (Rotary Position Encoding)

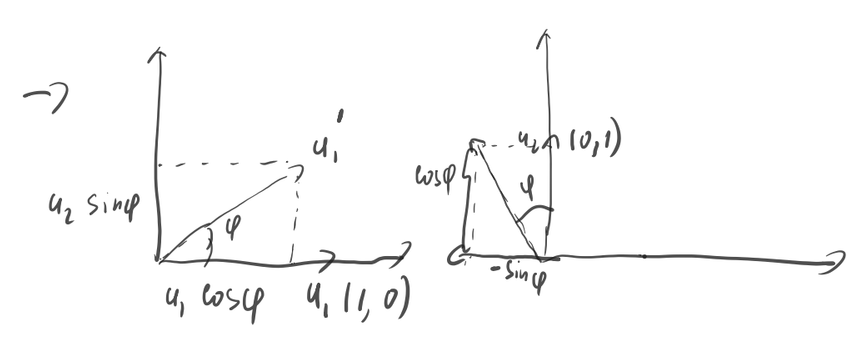
$$\boldsymbol{q}_m^\top \boldsymbol{k}_n = (\boldsymbol{R}^d_{\Theta,m}\boldsymbol{W}_q\boldsymbol{x}_m)^\top(\boldsymbol{R}^d_{\Theta,n}\boldsymbol{W}_k\boldsymbol{x}_n) = \underbrace{\boldsymbol{x}^\top\boldsymbol{W}_q}_{Q} R^d_{\Theta,n-m}\underbrace{\boldsymbol{W}_k\boldsymbol{x}_n}_{K^T}$$

$$\boldsymbol{R}^d_{\Theta,m} = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

$x^T W_q = Q$; $\overbrace{Q R}\, K^T = Q' K^T$

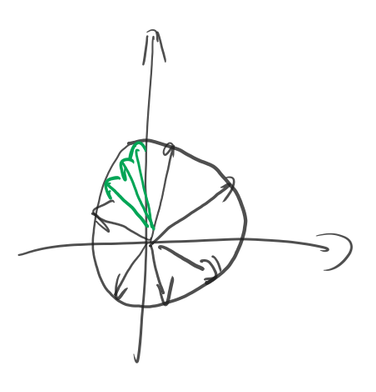

u'
u
u_2
φ
u_1'
$u_1 \cos\varphi$

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ u_2 \end{bmatrix}$$

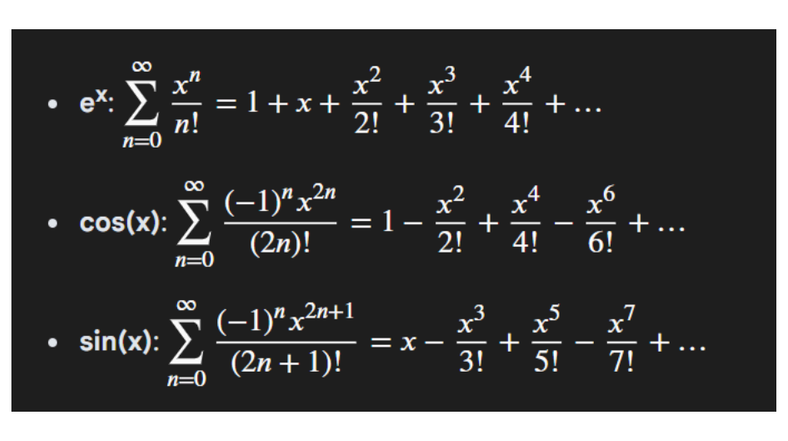

u_1'
$u_2 \sin\varphi$
φ
$u_1 \cos\varphi$
$u_1 (1,0)$
$\cos\varphi$
$u_2 (0,1)$
φ
$-\sin\varphi$

$$\theta_{p,j} = p\cdot\omega_j = p\cdot\frac{1}{\text{base}^{j/d}}$$

$q,\ R = (\overset{\text{complex}}{\underbrace{q_0, q_1}}, \underbrace{q_2, q_3}, \ldots)\ R =$



$= q_0 \cos m\theta_1 + q_1 \sin m\theta_1 \quad -q_0 \sin m\theta_1 + q_1 \cos m\theta_1$



- e^x: $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$
- $\cos(x)$: $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$
- $\sin(x)$: $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

• $z = r e^{i\theta_1}$

$z' = z \cdot e^{i\theta_2} = r e^{i(\theta_1+\theta_2)}$

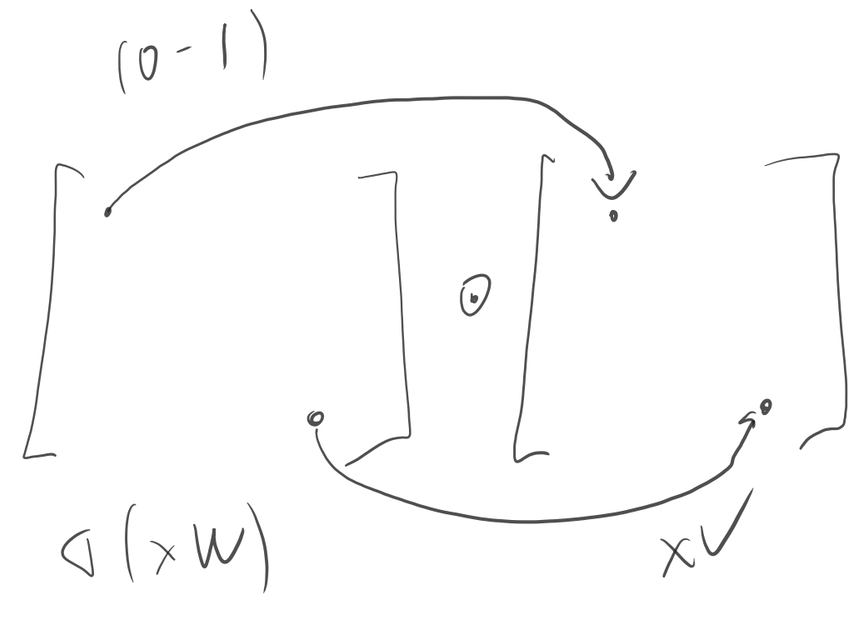
• Non-vanilla FFN

$FFN = \sigma(\underbrace{xW_1 + b_1}) W_2 + b_2$

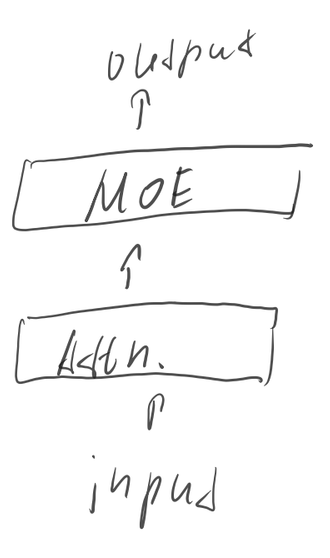
$$\begin{aligned} \text{FFN}_{\text{GLU}}(x, W, V, W_2) &= (\sigma(xW) \otimes xV)W_2 \\ \text{FFN}_{\text{Bilinear}}(x, W, V, W_2) &= (xW \otimes xV)W_2 \\ \text{FFN}_{\text{ReGLU}}(x, W, V, W_2) &= (\max(0, xW) \otimes xV)W_2 \\ \text{FFN}_{\text{GEGLU}}(x, W, V, W_2) &= (\text{GELU}(xW) \otimes xV)W_2 \\ \text{FFN}_{\text{SwiGLU}}(x, W, V, W_2) &= (\text{Swish}_1(xW) \otimes xV)W_2 \end{aligned}$$

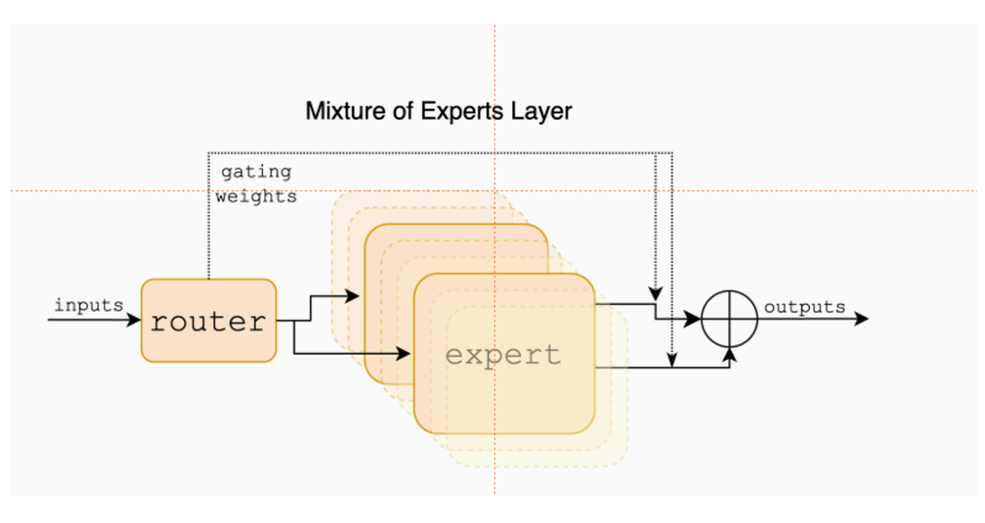
⊙ Hadamard's product

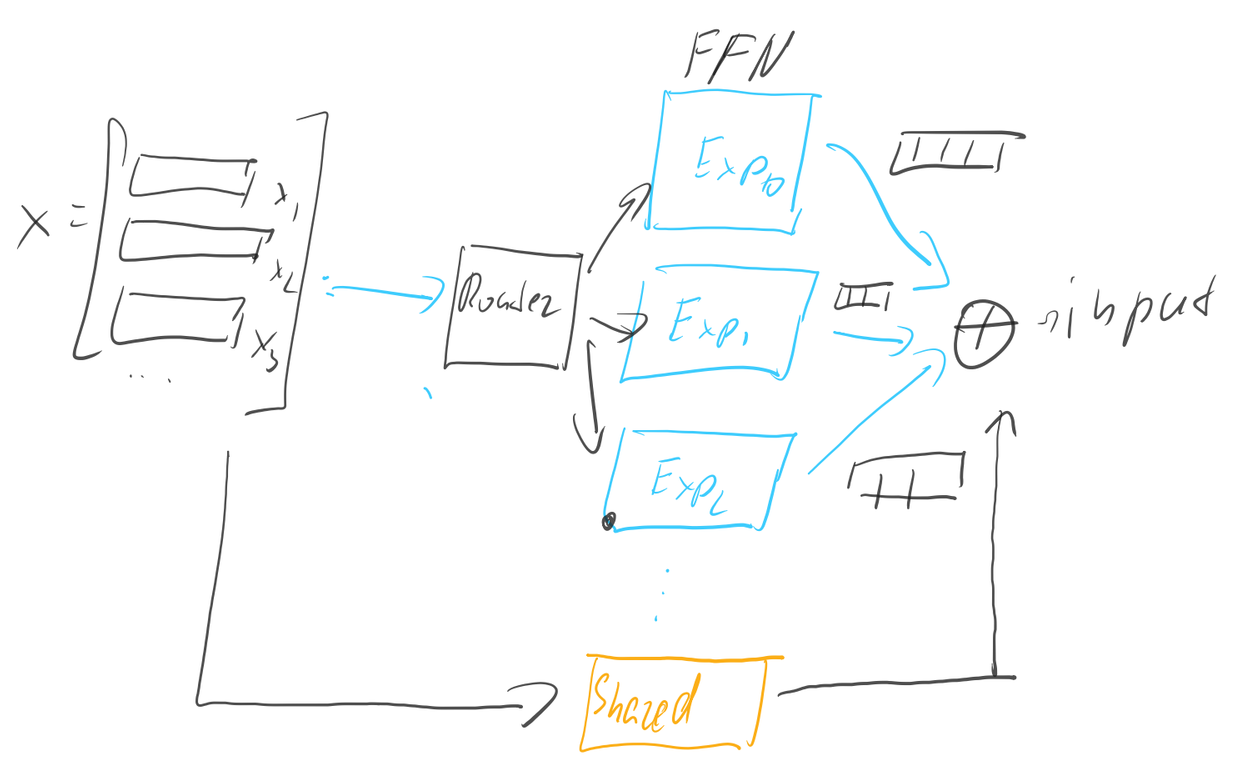
$\underbrace{\sigma(xW)}_{(0-1)} \odot xV$



• MoE (Mixture of Experts)


output
MoE
Attn.
input


Mixture of Experts Layer
gating weights
inputs
router
expert
outputs

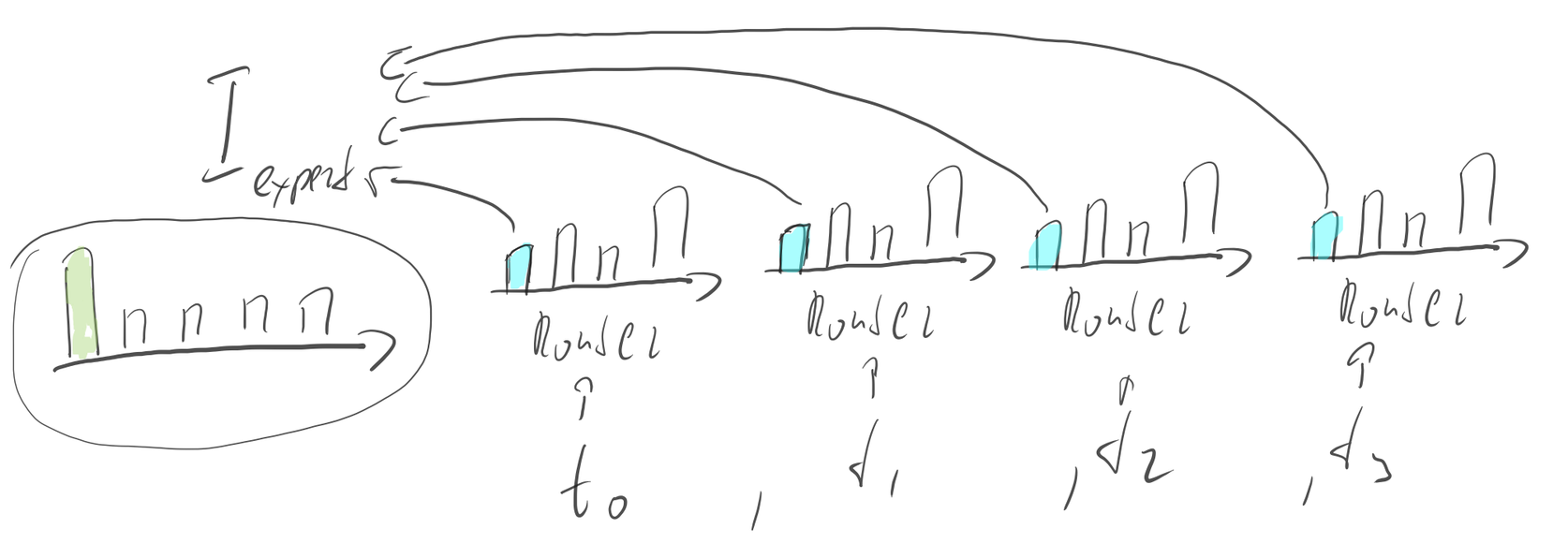

FFN
x =
x_1
x_2
x_3
Router
Exp_0
Exp_1
Exp_2
+ input
Shared

$\underset{d}{x_i}\ \underset{d\times\text{experts}}{W_{router}} \left([\ ] + \begin{bmatrix} -\inf \\ -\inf \\ 0 \\ 0 \end{bmatrix} \right) \to \text{softmax}\begin{pmatrix} -\inf \\ -\inf \\ a_2 \\ a_3 \end{pmatrix} \to \begin{pmatrix} 0 \\ 0 \\ 0,4 \\ 0,6 \end{pmatrix}$

Top-K (2)

• Auxiliary loss

$$CV = \frac{s.d(I_{expert})}{mean(I_{expert})}$$


I_{expert}
Router
Router
Router
Router
t_0 , t_1 , t_2 , t_3

$loss\_new = \underline{loss\_old} + \beta\ \underline{aux\_loss}$

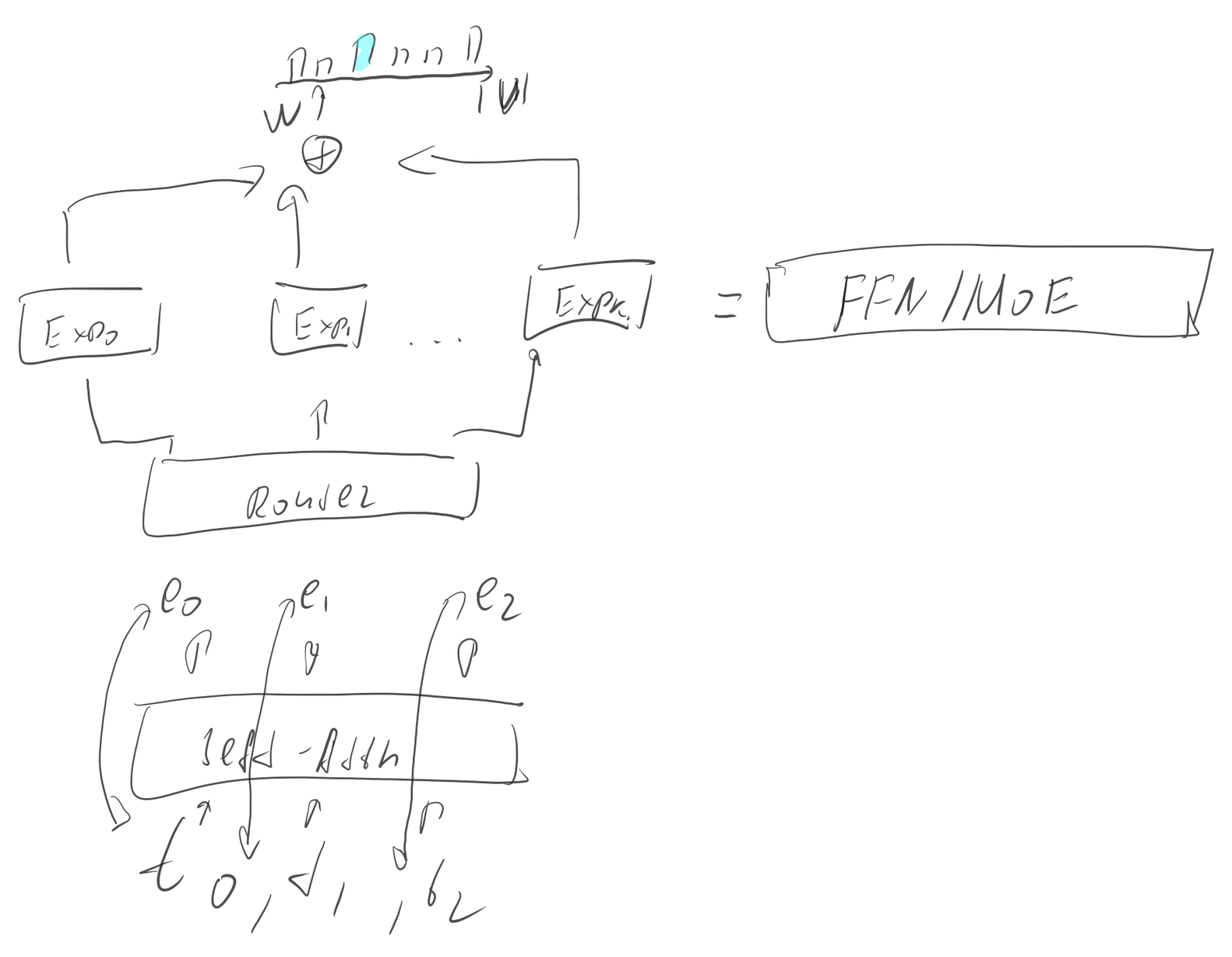
β → hyperparam.

• Gaussian noise

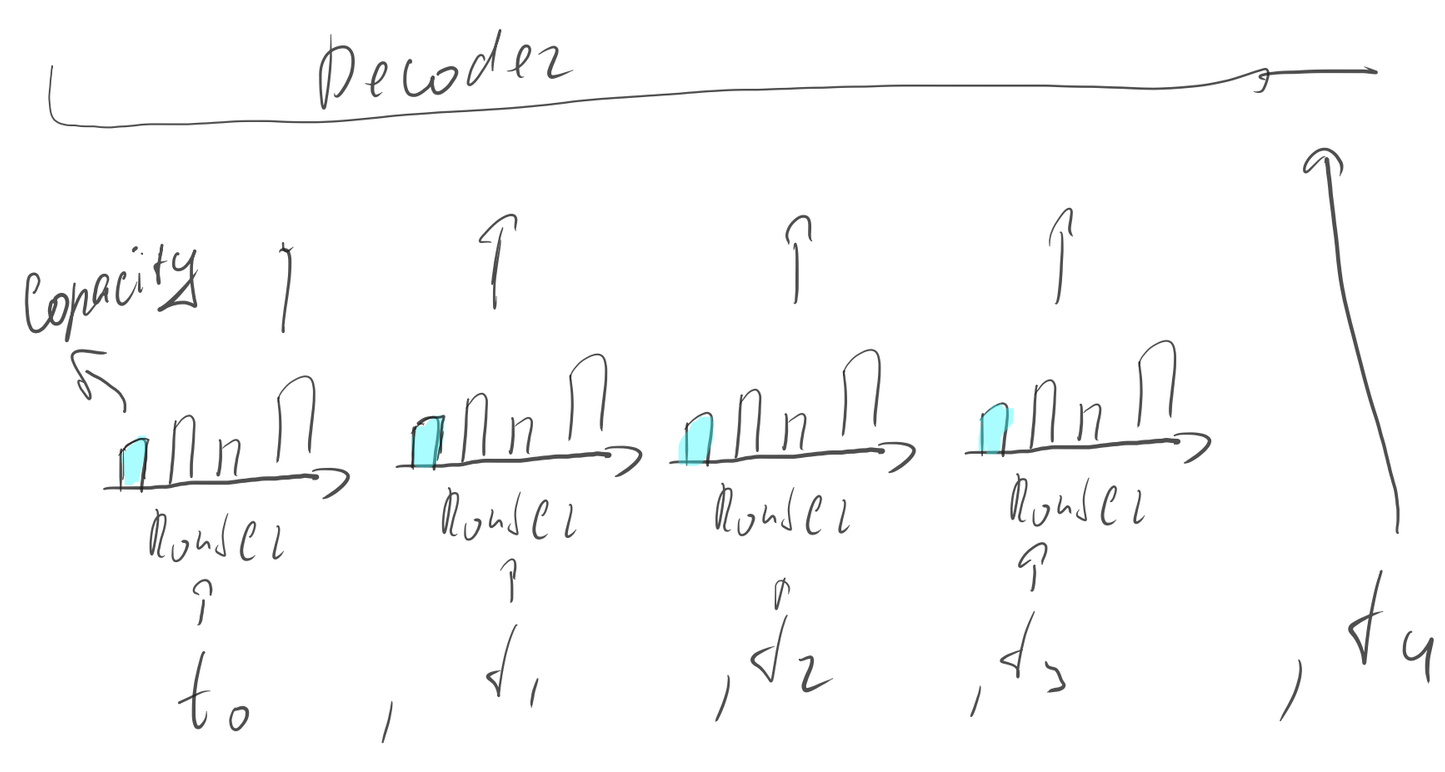
$N(0, \sigma(t)^2)$

$\to \underset{d}{[\ ]} \cdot \underset{d\times\text{experts}}{W}$

Exploitation vs. Exploration


W?
$|W|$
Exp_0
Exp_1
...
Exp_k
= FFN/MoE
Router
e_0
e_1
e_2
Self-Attn
t_0 , t_1 , t_2

• Expert capacity.


Decoder
Capacity
Router
Router
Router
Router
t_0 , t_1 , t_2 , t_3 , t_4

• Sparse Attention

$\underset{l\times d_h}{Q}\ \underset{d_h\times l}{K^T} \to O(l^2 d_h^2)$ - compute

$\to O(l^2)$ - memory

$X \to W^q \to \underset{l\times d_h}{Q}\ \underset{d_h\times d_e}{W}$ s.t. $d_h \gg d_e$

$\to W^k \to \ldots$

$\to W^v$